DRAFT — for Casey's review only. NOT FOR DISPATCH. (Keeper, 2026-08-25, v0.1)

Dear Grant,

I'm a retired computer scientist — Purdue, mid-1970s, systems and operating systems, some of the early UNIX networking stack — and for the past several years I've been working on a question with a research team, with every step published openly. I'm writing because the mathematics has become something I think you'd enjoy *looking at*, whatever you conclude about the physics.

The one-sentence version: take the rank-2 bounded symmetric domain $D\_IV^5 = SO(5,2)/[SO(5)\times SO(2)]$, and ask what its representation theory contains. The domain has five integer invariants — rank 2, and from it 3, 5, 6, 7 — all polynomial in the rank. Our claim, stated at carefully labeled evidence tiers, is that the *discrete* skeleton of the Standard Model — the gauge-group structure, the sixteen fermions of one generation with their hypercharges from a single rule, the chirality of the spectrum, three generations as rank+1 — is the linear algebra of this one object, with exactly one measured number taken as a ruler. The B_2 root system with its eight roots is, quite literally, the picture the whole thing grows from: give a child a ball and teach them to count.

Why I think this might interest you specifically: it is *visual and checkable* at every step. The repository has a one-command reproduction script (50 quantities against measurement, seconds to run), 2,000+ small computational "toys" that verify individual claims, and the geometry itself — a bounded domain, its boundary strata, a root diagram — is the kind of object your channel has made beautiful before.

And the part I'm proudest of, which is also the part you should probe hardest: we publish our failures with the same ceremony as our results. Our falsifier register has a "fired and lost" section. Just this week, we computed the unique *forced* geometric candidate for the fine-structure constant — the one with no step chosen to reach the answer — and it came out $8\pi^3/3$, which is not 1/137.036. We published that as a clean negative closing a fifty-year class of "α from volumes" proposals (Wyler's), including our own hopes for it. Every claim carries a tier label that tells a reader exactly what evidence stands behind it, and a one-page guide states what would change our minds about each.

There is no ask here beyond curiosity: the mathematics is on GitHub, the papers carry their own doubts on their faces, and if any of it strikes you as worth exploring — or debunking — both outcomes would honor the work. I'd be glad to point you at the shortest path in.

With admiration for what you've built, Casey Koons [repo link · Zenodo DOI 10.5281/zenodo.19454185]